# Structural certificates for the $A_{10}$ companion table

Private L054 finite application

**Theorem 1.** *Relative to the acquired complete ambient subgroup-class count, the complete local maximal-subgroup interfaces and the accepted structural registry, all 430 rows of the current $A_{10}$ companion table have the displayed structural descriptions. Each description asserts an actual decomposition, with the previously specified atomic conventions; it does not assert that an extension label determines a unique abstract group.*

**Source and scope.** The current table is attributed to Thomas Connor and Dimitri Leemans. All 21 native pages were directly inspected. Products and extensions are associated to the left unless the source supplies parentheses, as in the 2015 article's naming convention. For example, row 23 is $(A_6 \times 2) \cdot 2$; the dot is not replaced by a colon. Row 263 has $3^3$, whereas row 299 has the baseline integer 21. The dicyclic $Q_{12}$ and unsigned extraspecial $2^{1+4}$ conventions are acquired explicitly from the earlier structural proofs. Historical identity of these PDF bytes with the 2013 website is not asserted.

**Actual subgroup embeddings.** The previously saved group file has five proper representatives printed as alternating groups on explicit point supports. Those expressions are used only to propose candidate subgroups. We do not assume that a displayed name preserved the original producer's embedding.

The acquired complete subgroup-class count is 430. A new application of the accepted robust local-weight kernel verifies that the 430 candidates are subgroups of the actual natural $A_{10}$ and are pairwise nonconjugate in it. Consequently their classes exhaust all subgroup classes. The actual ambient model is checked against AlternatingGroup(10), and all candidate generators are then exported as literal ten-point permutations. This is the same counting and reconstruction argument accepted for $A_9, S_9$; no ambient subgroup enumeration is repeated.

For each candidate $H$, an acquired complete local maximal-subgroup interface is applied to a verified faithful smaller action. Duplicate local classes are removed by actual conjugacy, and inverse transport returns maximal subgroups of $H$. A local class of $K < H$ contributes $[H : N_H(K)]$ children. Contributions with the same ambient class are added. Writing $c_H, c_K$ for the ambient class lengths, the reverse weight is $ac_H/c_K$ by double counting; integrality and both class-length bounds are checked. The resulting 1592 weighted edges and all 184900 ordered source row pairs agree with the saved source bijection. In particular, the actual reconstructed subgroups have the correct source assignments.

**Literal structural verification.** The application acquires 286 concrete registry seeds and proposes 21 new models, with acyclic references to factors. A reused structure is transported by an explicit identity-preserving bijection which respects right multiplication by each ordered generator. Induction on generator words proves it is an isomorphism, the previously accepted generator-edge criterion. Every occurrence is joined to its exact embedded subgroup, not merely to another group of the same order.

For a split description the verifier checks $N \triangleleft H$, identifies $N$ and a subgroup $K$, verifies $N \cap K = 1$ and $|N||K| = |H|$, and hence proves $H = NK$. A colon includes a noncommuting pair of factor generators. For a direct product, all factor generators commute; then the factors commute elementwise. All subgroup, normality, product and commutation conditions are checked by literal finite permutation arithmetic.

There are twelve new split models, five direct models and four nonsplit models. Each new nonsplit model has quotient of order two. For such an extension $N \triangleleft H$, a complement exists if and only if some $x \in H \setminus N$ is an involution: the forward implication takes its nonidentity complement element; the reverse uses $\langle x \rangle$ and $[H : N] = 2$. The verifier checks all 8208 elements outside the four normal factors and finds none of order two. This is an exhaustive positive obstruction certificate, not a failed software complement search. The inherited normal and factor identifications are checked by the same literal maps.

Row 1 acquires the exact completed numerical construction $G :=$ AlternatingGroup(10) and the new ambient equality check. Its assigned subgroup has full order in $G$, so it is $G$ itself. The named constructor convention is recorded explicitly; the full ambient group is not expanded again for a redundant structural map.

**Validation and boundaries.** The embedding and source checks completed in 32.68 seconds. The single bounded witness pass returned all 429 proper-row witnesses in 29.01 seconds, with empty error output. These are proposal receipts; the independent literal verifier checks all model closures, factor maps and source joins. The verification checks 467 maps and 1948088 generator edges. Ordinary and optimized modes must return identical results. Controls damage actual generators, row assignments, source syntax, dots, degrees, input hashes, class lengths, weights, edges and bijections, and test a genuine split extension against the nonsplitting criterion.

All earlier mathematical criteria and their authorship remain acquired. No new classification algorithm, unique interpretation of extension labels, historical byte-identity claim, whole-source admission, question result, formalization or measurement is asserted. This is one of the larger source-associated tables, all of which remain in the recovery contract.